```
import pandas as pd
import numpy as np
```

```
url = 'https://raw.githubusercontent.com/prasertcbs/basic-dataset/master/netflix_titles.csv'
print("Downloading dataset...")
```

```
Downloading dataset...
```

```
df = pd.read_csv(url)
print("Original Shape:", df.shape)
```

```
Original Shape: (6234, 12)
```

```
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')
```

```
df.drop_duplicates(inplace=True)
print("Shape after dropping duplicates:", df.shape)
```

```
Shape after dropping duplicates: (6234, 12)
```

```
categorical_cols = ['type', 'title', 'director', 'country', 'rating']
for col in categorical_cols:
    df[col] = df[col].str.strip()
```

```
df['type'] = df['type'].str.title()
```

```
df['date_added'] = df['date_added'].str.strip()
df['date_added'] = pd.to_datetime(df['date_added'], format='mixed', errors='coerce')
```

```
print("\nMissing values before cleaning:\n", df.isnull().sum())
```

```

Missing values before cleaning:
 show_id           0
type              0
title             0
director       1969
cast            570
country         476
date_added       11
release_year      0
rating           10
duration          0
listed_in         0
description       0
dtype: int64
```

```
df['director'].fillna('Unknown Director', inplace=True)
df['cast'].fillna('Unknown Cast', inplace=True)
```

```
/tmp/ipykernel_2814/282196586.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through ch
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['director'].fillna('Unknown Director', inplace=True)
/tmp/ipykernel_2814/282196586.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through ch
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['cast'].fillna('Unknown Cast', inplace=True)
```

```
mode_country = df['country'].mode()[0]
df['country'].fillna(mode_country, inplace=True)
```

```
/tmp/ipykernel_2814/3401485373.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['country'].fillna(mode_country, inplace=True)
```

```
df.dropna(subset=['date_added', 'rating'], inplace=True)
```

```
df['release_year'] = df['release_year'].astype(int)
```

```
df['duration_num'] = df['duration'].str.extract('(\d+)').astype(float)
df['duration_num'].fillna(df['duration_num'].median(), inplace=True)
df['duration_num'] = df['duration_num'].astype(int)
```

```
<>:1: SyntaxWarning: invalid escape sequence '\d'
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/tmp/ipykernel_2814/444565257.py:1: SyntaxWarning: invalid escape sequence '\d'
  df['duration_num'] = df['duration'].str.extract('(\d+)').astype(float)
/tmp/ipykernel_2814/444565257.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through ch
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['duration_num'].fillna(df['duration_num'].median(), inplace=True)
```

```
print("\nFinal Data Types:\n", df.dtypes)
print("\nMissing values after cleaning:\n", df.isnull().sum())
```

```

Final Data Types:
 show_id                  int64
type                    object
title                   object
director                object
cast                    object
country                 object
date_added      datetime64[ns]
release_year             int64
rating                  object
duration                object
listed_in               object
description             object
duration_num             int64
dtype: object

Missing values after cleaning:
 show_id         0
type            0
title           0
director        0
cast            0
country         0
date_added      0
release_year    0
rating          0
duration        0
listed_in       0
description     0
duration_num    0
dtype: int64
```

```
df.to_csv('cleaned_netflix_dataset.csv', index=False)
print("\nDataset successfully cleaned and saved to Colab workspace as 'cleaned_netflix_dataset.csv'!")
```

```

Dataset successfully cleaned and saved to Colab workspace as 'cleaned_netflix_dataset.csv'!
```